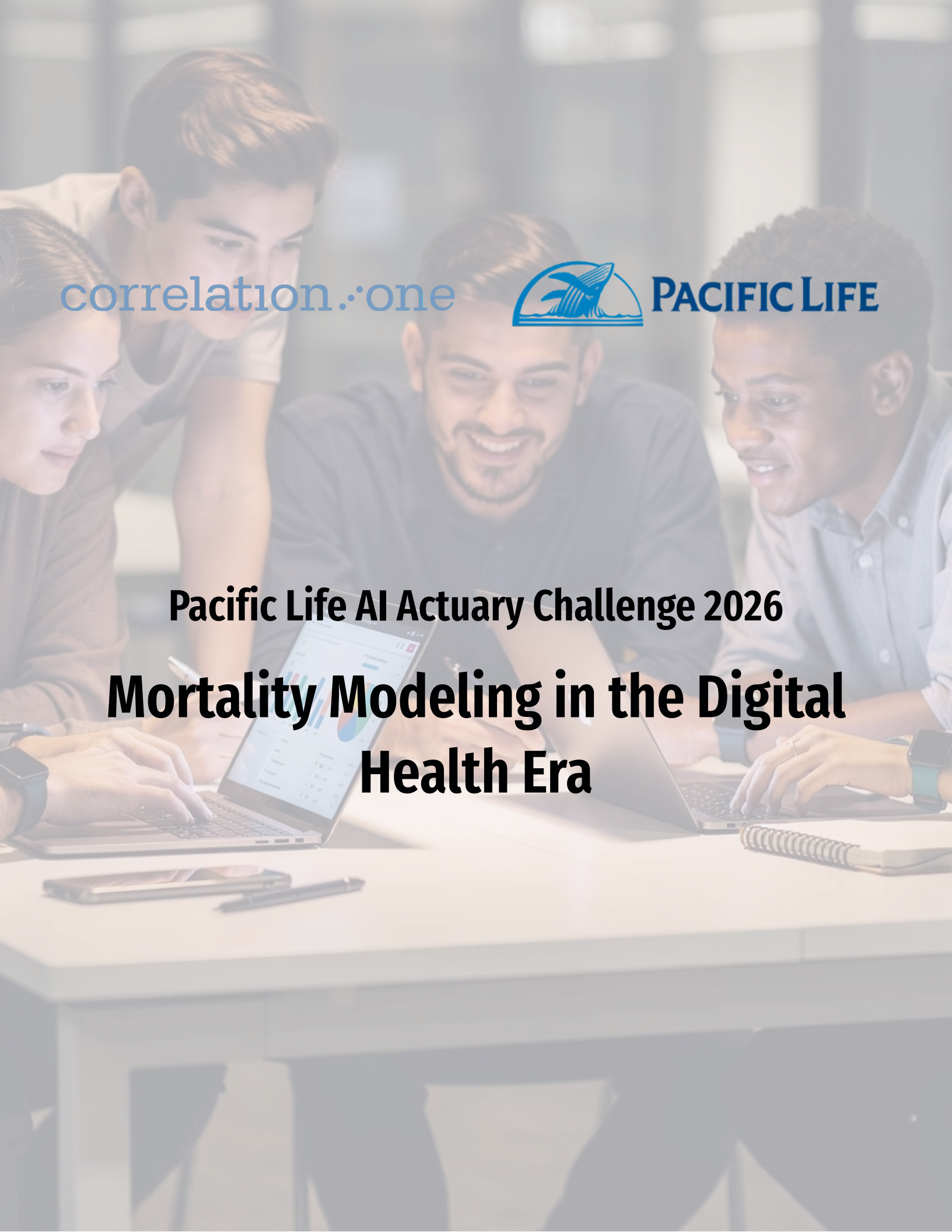

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PACIFIC LIFE

Pacific Life AI Actuary Challenge 2026

Mortality Modeling in the Digital Health Era

Problem Statement

Background

Over the past decade, consumer wearable devices have evolved from basic step counters into sophisticated health monitoring systems capable of detecting subtle physiological signals. Features such as irregular rhythm alerts, oxygen saturation variability, sleep quality tracking, and continuous heart rate monitoring have created new opportunities to identify elevated health risks earlier than traditional clinical pathways.

For life insurers, this raises a practical actuarial question: does wearable access and engagement provide reliable, decision-relevant information that improves long-term mortality risk assessment? If wearable data provides meaningful incremental signals, it may influence underwriting segmentation, long-term mortality assumptions, and product strategy. If it does not, or if the observed signal is primarily driven by selection effects, it could introduce bias and distortion into pricing and risk management.

Interpreting wearable signals is inherently complex. Individuals who adopt and actively engage with wearable devices may differ systematically from those who do not. Earlier diagnosis may reflect increased monitoring rather than improved health. Apparent mortality reductions may reflect confounding, treatment access, or behavioral differences rather than true causal impact.

As life insurers adopt AI-driven risk assessment tools, the challenge is not only predictive performance, but ensuring that new data sources improve decisions in a way that is stable, interpretable, and actuarially defensible.

This competition is designed to evaluate whether wearable data can be incorporated into actuarial mortality modeling in a way that is technically credible, interpretable, and actuarially defensible.

Scenario

You are an actuarial intern at a life insurance company that is modernizing its underwriting and pricing capabilities through AI-enabled risk assessment systems. As part of this effort, the company is exploring whether high-frequency wearable health data should be incorporated into its mortality modeling framework.

Advances in AI and data infrastructure now make it possible to process complex, longitudinal signals such as heart rate variability, sleep metrics, and activity trends. However, the availability of more data does not automatically translate into better actuarial decisions. New data sources must demonstrate measurable value, stability, interpretability, and fairness before being integrated into underwriting or pricing assumptions.

You have been given access to a retrospective 10-year mortality dataset for a population, including traditional underwriting variables and wearable-derived metrics. The retrospective study spans the years of 2016-2017 for wearable monitoring with clinical follow-ups occurring between 2018-2025.

Your task is to build a mortality risk model and evaluate the role wearable data plays within it. But beyond predictive performance, you are expected to address the broader actuarial question:

> Does wearable access and engagement provide meaningful, reliable incremental value for mortality risk assessment beyond traditional underwriting variables?

As well, you may pose and investigate additional analytical questions, provided they remain directly aligned with the overarching goal of assessing the impact of wearable health devices on mortality risk. Extensions that meaningfully deepen insight into mortality dynamics, causal mechanisms, or pricing implications are encouraged.

Submission Requirements

Teams must submit five items by the final deadline to **competitionsubmissions@correlation-one.com**:

1) Model Artifact (required)

Teams must submit a reproducible model artifact that can generate predicted 10-year mortality probabilities for the evaluation dataset. This artifact is used to score predictive performance on a hidden test set. The artifact should be stored as a .pkl file and named:

team_[n]_model_artifact.pkl, where [n] is your team number

The purpose of the artifact is to ensure transparency and integrity of results — not to reward complex pipelines. Simple, interpretable approaches are encouraged, provided they are well-validated and actuarially defensible.

2) Scoring Key

Using your model artifact, you must produce a scoring key that outputs a csv in the form of:

team_[n]_scoring_key.csv, where [n] is your team number

Columns:

- **ID** - Unique participant identifier from the test sets provided
- **predicted_probability** - The predicted probability (a continuous value between 0.0 and 1.0) that the participant died during the study period using provided test set data. Judges will use this to compute the log-loss.

Judges will compute the log loss using this .csv file on behalf of the team.

3) Code

Please submit all code and transformed data used to generate your model artifact and analyze it. We ask that you .zip them and submit it in the following format:

team_[n]_code.zip, where [n] is your team number

If the .zip is too large for an email submission, please submit to us using a publicly available cloud storage link (e.g. google drive).

Submission Requirements (continued)

4) Final Presentation Deck + AI Use Disclosure

Teams must submit a slide deck in PDF format (recommended 8–10 slides + appendix with technical details). This is the primary judged deliverable.

The presentation must include a dedicated slide on AI Use describing:

- Whether and how generative AI tools were used
- How AI-generated outputs were validated
- Any notable errors, limitations, or corrections identified

This disclosure must be included as a single slide within the final presentation.

This file should be named:

team_[n]_presentation.pdf, where [n] is your team number

5) Video Presentation (7 minutes maximum)

Teams must also submit a recorded video presentation (maximum 7 minutes) summarizing their analysis and recommendation.

The video should clearly communicate:

- The core actuarial question and hypothesis
- The modeling approach and key findings
- Practical implications for underwriting
- The final recommendation

The video should align with the submitted slide deck and reflect the same conclusions.

Each member of the team should have a part in the video presentation.

This file should be named:

team_[n]_video_presentation.(mp4/mov), where [n] is your team number

Presentation Deck Requirements

The deck should address:

Actuarial Question and Hypothesis

The team's framing of the actuarial question and hypothesis

Wearable Data Incremental Value

Whether wearable data provides incremental value beyond traditional underwriting variables

Key Limitations and Risks

Key limitations, selection effects, and confounding risks

Validation and Reasonableness Checks

Validation and reasonableness checks (including calibration and robustness)

Stress Tests or Scenario Analyses

Stress tests or scenario analyses demonstrating how conclusions may change under different assumptions

Clear Recommendation for Workflow

A clear recommendation for how wearable data should be used in an AI-enabled underwriting workflow (including the option to recommend not using it)

Notes on Expectations

Teams are not expected to build production-grade software. The evaluation artifact is intended to support reproducibility and objective scoring.

Teams may use any reasonable modeling approach. However, models that are overly complex without clear interpretability, validation, and actuarial justification are unlikely to score well in the presentation evaluation.

The strongest submissions will demonstrate actuarial judgment: clear reasoning, appropriate skepticism, and decision-ready recommendations grounded in evidence.

> **The top three-ranked teams will be invited to present live to Pacific Life judges. This final live presentation will determine the ultimate competition rankings.**

Overall Evaluation Structure

Submissions are evaluated using two components:

20%

Predictive Performance

Models are evaluated on a hidden test set using a proper scoring rule for probabilistic mortality prediction. Predictive performance is included to ensure baseline technical competence and to reward models that produce accurate, well-calibrated probability estimates.

80%

Actuarial Presentation

Teams are evaluated on the quality of actuarial reasoning, interpretation, and communication. The goal of this competition is to identify students who can reason about mortality risk and clearly communicate findings—not simply create the most predictive model.

Generative AI Use

Pacific Life recognizes that generative AI tools are increasingly used in modern actuarial and analytical workflows, particularly for drafting documentation, structuring analysis, and accelerating exploration of complex problems. Participants may use generative AI tools (e.g., ChatGPT or similar) as part of their work in this competition, provided usage is responsible, transparent, and does not replace actuarial reasoning.

Permitted Uses

Teams may use generative AI to support tasks such as:

- **Structuring & Synthesis**: Summarizing exploratory findings and structuring written deliverables
- **Communication**: Drafting or refining presentation language and executive-ready communication
- **Technical Brainstorming**: Ideating modeling approaches or validation strategies
- **Coding Assistance**: Helping with debugging and code generation
- **Risk Identification**: Identifying potential limitations in your work

Not Permitted Uses

To maintain the integrity of the competition, the following are strictly prohibited:

- **Blind Submission**: Submitting AI-generated outputs without thorough review, validation, and manual refinement.
- **Fabrication**: Using AI to create "hallucinated" evidence, citations, analysis steps, or conclusions.
- **Reverse Engineering**: Using AI to attempt to reverse-engineer the competition datasets

Generative AI Use

Accountability and Validation

Teams are responsible for all work submitted, regardless of whether AI tools were used. Any AI-assisted content must be reviewed for accuracy, hallucinations, and consistency with the team's actual analysis. The ability to validate AI-generated material is considered part of responsible AI-enabled actuarial practice.

Disclosure Requirement

In the presentation deck, teams must include a dedicated AI Disclosure Slide(s). Your disclosure should:

- **Delineate Responsibility**: Clearly showcase where AI provided a "starting spark" versus where human actuarial reasoning took over to refine, validate, or override the AI's suggestion.
- **The "Why," not just the "How"**: Don't just list the prompts used. Explain why the team chose to accept or reject specific AI-generated suggestions based on actuarial principles.
- **Validation Process**: Briefly describe the validation steps taken to ensure AI-assisted code or logic met the professional standards required for the problem set.

AI is a supplement to, not a substitute for, your professional judgment. Teams must demonstrate a primary understanding of their chosen methodologies and remain fully responsible for the interpretation and defense of all results.

Data Overview

Participant Health Profile Dataset

This is data containing participant-level demographic, behavioral, and clinical risk factors designed for mortality modeling and risk analysis.

Column	Description
ID	Unique participant identifier. Note: that train and test set IDs are independent namespaces and should not be concatenated.
wearable	An indicator showing whether the participant used a wearable device. 0=no wearable device or non-engaging wearable user and 1=Has wearable device.
engagement_level	Categorical measure of how actively the participant engages with the wearable. 0 =no engagement and 1–5 = increasing engagement tiers.
age	Participant age in years.
age_bin	Age is grouped into categorical bins for stratified analysis and modeling.
sex	Biological sex of the participant.
height	Participant height in centimeters
weight	Participant weight in kilograms
bmi	Body Mass Index is calculated from height and weight.
family_medical_diabetes	A categorical variable that the participant has a family history of diabetes. 0 for no history, 1 for sibling(s) affected, and 2 for parent(s) affected

Participant Health Profile Dataset (continued)

Column	Description
family_medical_diabetes_type	Type of diabetes reported in the family history, such as type 1 or type 2.
family_medical_cancer	An indicator that the participant has a family history of cancer. 0= No history and 1= immediate family history of cancer
family_medical_dementia	A categorical variable that the participant has a family history of dementia. 0 for no history, 1 for parent(s) affected, and 2 for early onset family history.
family_medical_heart_disease	A categorical variable that the participant has a family history of heart disease. 0 for no history, 1 for sibling(s) affected, and 2 for parent(s) affected
smoker	An indicator that the participant is a current smoker. 0=not a smoker and 1 = a smoker.
weekly_exercise	Typical amount of exercise performed per week in days/week.
alcohol_consumption_frequency	Frequency of alcohol consumption, measured as a categorical variable. 0=never, 1=<monthly, 2=monthly, 3=weekly, 4=2–3×/week, 5=unassigned, 6=daily.
diagnosed_diabetes	An indicator that diabetes has been formally diagnosed by a clinician. 0=not diagnosed with diabetes and 1=diagnosed with diabetes.
diagnosed_diabetes_type	Type of diabetes that has been clinically diagnosed.
diagnosed_lung_cancer	An indicator that lung cancer has been clinically diagnosed. 0=not diagnosed with lung cancer and 1=diagnosed with lung cancer.

Participant Health Profile Dataset (continued)

Column	Description
diagnosed_other_cancer	An indicator that a non-specified type cancer has been clinically diagnosed. 0=not diagnosed with cancer and 1=diagnosed with non-specified type cancer.
diagnosed_sleep_apnea	An indicator that sleep apnea has been clinically diagnosed. 0=not diagnosed with sleep apnea and 1=diagnosed with sleep apnea.
diagnosed_afib	An indicator that atrial fibrillation has been clinically diagnosed. 0=not diagnosed with AFib and 1=diagnosed with AFib.
diagnosed_hypertension	An indicator that hypertension has been clinically diagnosed. 0=not diagnosed with hypertension and 1=diagnosed with hypertension.
diagnosed_dementia	An indicator that dementia has been clinically diagnosed. 0=not diagnosed with dementia and 1=diagnosed with dementia.
outcome	Binary indicator showing whether the participant died during the study period. 0= Did not die in the study period and 1= Died during the study period.

Wearable Time Series Dataset

This dataset contains daily wearable device measurements for each participant across the study period. Measurements taken during 2016–2017. **Note: wearable device adoption is self-selected rather than randomly assigned.**

Column	Description
ID	Unique participant (user) identifier for the wearable device owner. Note: that train and test set IDs are independent namespaces and should not be concatenated.
timestamp	Date and time when the daily wearable measurements were recorded.
day_of_week	Day of the week corresponding to the timestamp.
snoring	An indicator or score representing whether snoring was detected during sleep that night. 0=No snoring detected that night and 1=Snoring detected.
n_wakeups	Number of times the participant woke up during the night.
sleep_duration	Total sleep duration for the night. Measured in hours.
sleep_light	Time spent in light sleep during the night. Measured in hours.
sleep_deep	Time spent in deep sleep during the night. Measured in hours.
sleep_rem	Time spent in REM sleep during the night. Measured in hours.
afib_daily	Daily indicator for atrial fibrillation detection from the wearable device. 0=No AFib detected that day and 1= AFib detected that day.
exercise	An indicator showing whether any exercise activity was recorded that day. 0=No exercise recorded that day and 1=Exercise recorded.

Wearable Time Series Dataset (continued)

Column	Description
exercise_code	Encoded value representing the type of exercise performed. 0 = running, 1=lifting, 2=swimming, 3=basketball, 4=tennis and 5=rock-climbing.
exercise_type	Human-readable exercise category. Values include basketball, lifting, rock-climbing, running, swimming & tennis.
exercise_duration	Total duration of exercise for that day. Measured in minutes.
steps	Total number of steps recorded during the day.
SpO2_U	Upper (maximum) blood oxygen saturation (SpO_2) value recorded for the day. Measured in % oxygen saturation.
Sp02_L	Lower (minimum) blood oxygen saturation (SpO_2) value recorded for the day. Measured in % oxygen saturation.
RHR_U	Upper (maximum) resting heart rate value recorded for the day. Measured in bpm.
RHR_L	Lower (minimum) resting heart rate value recorded for the day. Measured in bpm.
MaxHR	The maximum heart rate recorded during the day. Measured in bpm.
dominant_hand	The participant's dominant hand (left/right).
state	U.S. state of the participant's primary residence.
region	Broader geographic region of the participant's residence (e.g., Northeast, South).

Wearable Time Series Dataset (continued)

Column	Description
hospital_state	U.S. state where the participant's primary hospital is located.
household_size	Number of people residing in the participant's household.
altitude	Altitude (in feet) of the participant's location.
os	Operating system of the participant's wearable device (e.g., iOS, Android).
app_version	Version of the wearable app installed on the participant's device.
battery_level	Battery level percentage (0-100%) of the wearable device at time of the daily recording.
skin_temp	Skin temperature (°C) reading captured by the wearable sensor.
atmospheric_temp_lower	Lower bound of ambient atmospheric temperature recorded on that day (°C).
atmospheric_temp_upper	Upper bound of ambient atmospheric temperature recorded on that day (°C).
app_crash	Indicator of whether the wearable app crashed during the recording period for that day. 0=No app crash and 1=App crashed during recording.

Diagnosis Dataset

This dataset contains medical diagnosis records for patients between 2018-2025. Each row represents a diagnosis event, including the patient ID, the hospital or clinic where the visit occurred, the healthcare provider involved, the date of the visit, the medical condition category, and the specific diagnosis given. It is structured for tracking and analyzing patient diagnoses across providers and facilities.

Column Name	Description
ID	Unique identifier for the patient or individual associated with the diagnosis record. Note: that train and test set IDs are independent namespaces and should not be concatenated.
HOSPITAL_CLINIC	Name of the hospital or clinic where the diagnosis or visit occurred.
HCP	Healthcare Provider. The physician or medical professional responsible for the diagnosis or visit.
date	Date when the diagnosis was recorded or when the patient visit occurred.
diagnosis	Specific diagnosis given to the patient. This is unstructured clinical text.

Dataset Artifacts

You will be provided the following .csv files:

- **train_participant.csv** — 20,000 rows × 26 columns
- **test_participant.csv** — 20,000 rows × 25 columns (target column excluded)
- **train_wearable.csv** — 507,864 rows × 33 columns
- **test_wearable.csv** — 508,387 rows × 33 columns
- **train_diagnosis.csv** — 11,537 rows × 5 columns
- **test_diagnosis.csv** — 11,487 rows × 5 columns